Security Operations Centre
Incident Investigation Report

Incident Reference: Scenario_2 / Windows_VM Discovery Activity

Report Date: 27 October 2026

Classification: Internal Lab Analysis

1. Executive Summary

Analysis of supplied Wazuh alerts and Sysmon (Event ID 1) plus supporting Windows Security events from host DESKTOP-5OG7C28 (agent Windows_VM, 192.168.232.130) reveals a short burst of local discovery activity executed under the interactive user account 'Darragh Casey'. A PowerShell process (PID 12276, High integrity) spawned multiple instances of hostname.exe and whoami.exe, followed by a cmd.exe instance that executed a chained command performing user enumeration, directory listing of C:\Users, Credential Manager enumeration via cmdkey.exe /list, and local group enumeration via net.exe / net1.exe. Corresponding Wazuh rules (92052, 92036, 92031, 92032) fired on these process creations. Security Event ID 5379 confirms credentials were enumerated, and Event ID 4799 confirms local group membership enumeration. No evidence of network activity, persistence mechanisms, file downloads, or further post-discovery actions is present in the supplied telemetry. The activity is consistent with reconnaissance typical of an initial foothold or a scripted test. Given the isolated laboratory context and the presence of a related PowerShell script file (Test_2.ps1), the overall assessment is that this represents deliberate discovery activity rather than random noise. Confidence: High on the observed facts; Medium on broader intent pending additional context.

2. Scope of Activity

The supplied evidence covers a narrow time window on 27 October 2026, primarily between approximately 11:29 UTC and 11:34 UTC (local timestamps in the +0100 zone). Activity is confined to a single Windows host (DESKTOP-5OG7C28) under the local user Darragh Casey. Observed processes include standard Windows utilities (hostname.exe, whoami.exe, cmd.exe, net.exe, net1.exe, cmdkey.exe) parented by powershell.exe, plus an earlier Notepad.exe instance that opened a .ps1 file. System-level processes (svchost.exe, taskhostw.exe) and routine registry updates appear but are unrelated to the discovery chain. No cross-host activity, external network connections, or long-running processes beyond the brief command executions are evidenced. The scope is limited to local account, group, credential, and basic system discovery.

3. Detailed Investigation

Observed Facts: At 11:29:36 UTC, explorer.exe launched Notepad.exe (PID 8532, Medium integrity) with the command line opening 'C:\Users\Darragh Casey\Desktop\S_2\Test_2.ps1'. Subsequent AppCompatFlags registry value sets by svchost.exe recorded the Notepad execution. At 11:34:41 UTC, powershell.exe (PID 12276, High integrity, parent command line consisting only of the executable path) began spawning child processes. Four HOSTNAME.EXE and two whoami.exe instances were created in quick succession, with working directory C:\Users\Darragh Casey\Desktop\S_2\. At 11:34:42.466 UTC the same PowerShell spawned cmd.exe (PID 1832) with the exact command line: "cmd.exe" /c net user & dir c:\Users\ & cmdkey.exe /list & net localgroup "Users" & net localgroup. Working directory had shifted to the

user's Temp folder (C:\Users\DARRAG~1\AppData\Local\Temp\). This cmd.exe then parented: net.exe (PID 12068) running 'net user' which itself spawned net1.exe (PID 6584); cmdkey.exe (PID 7940) running 'cmdkey.exe /list'; net.exe (PID 8816) running 'net localgroup "Users"' which spawned net1.exe (PID 10484); and net.exe (PID 6504) running 'net localgroup' which spawned net1.exe (PID 4912). Windows Security Event 5379 (Credential Manager credentials were read – Enumerate Credentials) was logged at 11:34:42.569 UTC with client process ID matching the cmdkey.exe. Event 4799 (security-enabled local group membership enumerated for the Users group) was logged at 11:34:42.612 UTC with caller process net1.exe. Wazuh alerts of level 3–4 were generated for the abnormal parent of cmd.exe, net.exe started by cmd, and the discovery commands.

Inferences: The Notepad opening of Test_2.ps1 immediately prior to the PowerShell-driven discovery strongly suggests the user reviewed or prepared a script containing the observed commands, then executed it (or equivalent interactive commands) from PowerShell. The High integrity level indicates the PowerShell session was elevated. The chained command in cmd.exe is a classic one-liner for bulk discovery, common in both malicious recon scripts and administrative or training exercises. Absence of further events after the final net1.exe indicates the activity concluded after enumeration. The parent PowerShell command line lacking arguments is consistent with an interactive session or a script invoked in a manner not captured by the process create event (process create for PowerShell itself is not present in the archives).

4. Evidence Analysis

Primary evidence is Sysmon Event ID 1 (Process Create) records from both the Wazuh alerts (Scenario_2_Alerts.json) and the fuller archive (Scenario_2_Archives.json). Key supporting fields include Image, CommandLine, ParentImage, ParentCommandLine, ParentProcessId, User, IntegrityLevel, CurrentDirectory, and Hashes. The alerts specifically flag rule 92052 (cmd.exe started by abnormal process – PowerShell), multiple 92036 (net.exe started by cmd), 92031 (discovery activity – account discovery), and 92032 (suspicious cmd shell). Security Event 5379 directly corroborates the cmdkey.exe /list action by recording the credential enumeration operation. Event 4799 corroborates the net localgroup enumeration. Earlier Notepad process create and AppCompat registry events establish the presence and user interaction with Test_2.ps1. System events (svchost GPSvcGroup, taskhostw, logon type 5 for SYSTEM) are present but show no causal link to the user discovery chain and are assessed as background OS activity. No Sysmon Event ID 3 (network), 11 (file create), 12/13/14 (registry persistence), or 7 (image load) of interest appear in the supplied data, limiting visibility to process execution and the two Security events.

5. Timeline of Events

• 11:29:36 UTC – explorer.exe (PID 6780) starts Notepad.exe (PID 8532) opening Desktop\S_2\Test_2.ps1 (Medium integrity).

• 11:29:38 / 11:29:59 UTC – svchost.exe sets AppCompatFlags registry values for the Notepad.exe path.

• 11:30:08 UTC – System svchost.exe (GPSvcGroup) and related Security 4624/4672 for SYSTEM logon (unrelated background).

• 11:31:51 UTC – taskhostw.exe started under SYSTEM (unrelated).

• 11:34:41.717–11:34:42.169 UTC – powershell.exe (PID 12276, High) spawns multiple HOSTNAME.EXE and whoami.exe processes (CWD: Desktop\S_2).

• 11:34:42.466 UTC – powershell.exe spawns cmd.exe (PID 1832) with chained discovery command (CWD: Temp).

• 11:34:42.515–11:34:42.648 UTC – cmd.exe parents net.exe/net1.exe (net user; net localgroup "Users"; net localgroup) and cmdkey.exe /list.

• 11:34:42.569 UTC – Security Event 5379: Credential Manager credentials enumerated (client PID matching cmdkey).

• 11:34:42.612 UTC – Security Event 4799: Local group 'Users' membership enumerated by net1.exe.

• ≈12:34 local – Wazuh alerts generated for the above process creations.

6. MITRE ATT&CK; Mapping

T1059.003 – Command and Scripting Interpreter: Windows Command Shell
Evidence: cmd.exe spawned by powershell.exe with /c and a chained command line containing net, dir, and cmdkey. Multiple alerts explicitly map this technique.

T1087 – Account Discovery (and sub-techniques for Local Account)
Evidence: Execution of 'net user', 'net localgroup "Users"', 'net localgroup', and whoami.exe. Security Event 4799 confirms group membership enumeration. Alerts 92031 map T1087.

T1555.004 – Credentials from Password Stores: Windows Credential Manager
Evidence: cmdkey.exe /list executed, corroborated by Security Event 5379 'Credential Manager credentials were read' with Enumerate Credentials operation.

T1082 – System Information Discovery (supporting)
Evidence: Multiple hostname.exe executions from the same PowerShell process.

These mappings are directly supported by process command lines and correlating Security events. No techniques beyond Discovery and Execution are evidenced.

7. Overall Investigation Conclusion

The supplied telemetry documents a concise, scripted or interactive local discovery sequence executed by the elevated PowerShell process of user Darragh Casey. The commands systematically enumerated local accounts, groups, stored credentials, hostname, and the current user identity. All key actions are directly observed in Sysmon process creation records and corroborated by Windows Security events. The prior opening of Test_2.ps1 in Notepad provides a plausible source for the command set. No indicators of subsequent stages (persistence, lateral movement, data exfiltration, or C2) exist in the provided logs. In an isolated laboratory environment this activity is most consistent with a controlled test or training exercise of discovery techniques; however, the same pattern would be treated as suspicious reconnaissance in a production environment. **Observed facts** (process trees, command lines, event IDs, timestamps, user context) are of high confidence. **Inferences** regarding the exact invocation method of PowerShell and the precise content of Test_2.ps1 are of medium confidence, as the process create for PowerShell and the script body are not present. Overall investigation confidence: High.

End of Report

Generated from analysis of Scenario_2_Alerts.json and Scenario_2_Archives.json only. No external intelligence applied.